Name:Prathysha Lakshmi K Register No:250701546 Experiment Name:Outliers Detection

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import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

array = np.array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79,
61, 90, 54])

def outDetection(arr):
    Q1, Q3 = np.percentile(arr, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)

final_array = array[(array >= lr) & (array <= ur)]

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
sns.histplot(final_array, kde=False)

plt.subplot(1, 2, 2)
sns.kdeplot(final_array, fill=True)

plt.tight_layout()
plt.show()
```

